

Notes: Hebert (RFS, 2025)

Gender Stereotypes and Entrepreneur Financing

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1 One-sentence summary

This note summarizes Hebert (2025), which uses French administrative startup data, the SINE entrepreneur survey, firm registry records, tax files, and SME financing survey evidence to study whether female founders face a gender gap in early-stage equity financing, showing that the gap is concentrated in male-dominated sectors, only partly explained by founder and startup characteristics, accompanied by substitution from equity to bank debt among growth-oriented women, and validated by outcome tests in which VC-backed female-founded startups outperform comparable male-founded startups in male-dominated sectors.

2 Big picture

2.1 Research question

The paper asks whether women entrepreneurs have less access to external equity and venture capital, and whether this difference reflects observable differences in founder quality, project quality, entrepreneurial motivation, and financing demand, or whether it is consistent with context-dependent stereotypes in investor decision making.

The core question is not simply whether female-founded startups raise less VC. The more precise question is:

When male and female entrepreneurs look similar ex ante, start firms in the same sector, year, and local market, and display similar growth intentions, do they obtain the same type of financing?

2.2 Main answer

The answer is no for external equity. Female-founded startups are less likely to use external equity and VC than comparable male-founded startups. The gap is strongest in male-dominated sectors and disappears in female-dominated sectors. The author argues that this sector-specific pattern is consistent with context-dependent stereotypes rather than a simple average difference in startup quality or average demand for growth.

2.3 Key empirical facts

- Women found about 29% of new firms in the SINE sample, but only around 1.7% of female-founded startups use external equity compared with 2.9% of male-founded startups.
- Conditional on sector-year, county, founder characteristics, startup characteristics, initial capital, and motivations, female-founded startups are 22% less likely to use external equity and 36% less likely to receive VC.
- The gender gap is concentrated in male-dominated sectors: female-founded startups are 27% less likely to use external equity and 50% less likely to use VC in those sectors.
- Observable characteristics explain 42% of the average external-equity gender gap, but 80% of the gap in male-dominated sectors remains unexplained.
- Growth-oriented female entrepreneurs in male-dominated sectors rely more on bank loans, suggesting debt substitutes for missing equity financing.

- VC-backed female-founded startups in male-dominated sectors outperform comparable male-founded startups in short-run survival and revenue growth, consistent with a higher funding bar for women.

3 Incremental contribution of the paper

3.1 Contribution to entrepreneurial finance

The paper extends the entrepreneurial finance literature by moving beyond funded-startup data. Most VC datasets observe only firms that successfully raised external equity, which makes it difficult to study the counterfactual population of firms that might have sought or used equity but did not obtain it. This paper uses representative survey and administrative data on new firms, including unfunded firms, so it can examine the full population of startup financing choices.

3.2 Contribution to gender and discrimination research

The paper provides a context-dependent test of stereotypes. Instead of estimating a single average gender gap, it asks whether the gap varies with the gender composition of the sector. This is important because stereotypes often operate most strongly when a person belongs to a minority group in a domain associated with another group. The finding that the equity gap is concentrated in male-dominated sectors, but not in female-dominated sectors, is central to the paper’s interpretation.

3.3 Contribution to identification

The paper combines four complementary strategies:

1. rich controls for founder profile, startup type, initial capital, sector, local market, year, and motivations;
2. cross-sector heterogeneity based on gender congruence;
3. Oaxaca–Blinder decomposition to separate explained and unexplained components;
4. outcome tests comparing ex-post performance of funded male- and female-founded firms.

This layered strategy makes the argument more convincing than a simple regression of financing on founder gender.

3.4 Contribution to financing substitution

The paper shows that female entrepreneurs with growth-oriented profiles may substitute external equity with bank debt. This connects the gender gap in equity finance to capital structure and financing choice. The result also emphasizes that “not receiving VC” does not mean “not seeking growth.” Some women appear to finance growth with less suitable instruments because equity access is limited.

4 Data and sample construction

4.1 Core data sources

The paper combines several French data sources:

- **SINE survey:** a representative survey of entrepreneurs who founded firms in France in 2002, 2006, 2010, 2014, and 2018. It includes founder demographics, education, experience, motivations, financing sources, incorporation status, startup capital, and growth intentions.
- **Firm registry:** identifies location, industry, incorporation status, and sector gender composition.
- **Corporate tax files and payroll data:** provide performance outcomes such as survival, sales growth, employment growth, assets, and tangibility.
- **PME survey:** provides out-of-sample evidence on bank applications, guarantees, and reasons for rejection for growing SMEs.
- **Crunchbase France:** used in robustness and Internet Appendix analysis for equity deal types and intensive margins.

4.2 Key sample features

The main SINE sample contains 131,284 startups. About 71% are male-founded and 29% are female-founded. Male-dominated sectors contain 80% of new firms; female-dominated sectors contain 20%. A sector is female-dominated if at least 50% of newly created firms in the four-digit sector are female-founded.

5 Conceptual framework and empirical predictions

5.1 Statistical discrimination versus stereotypes

The paper distinguishes between three mechanisms:

- **Statistical discrimination:** investors use gender as a proxy for true average differences in expected project quality. If their beliefs are accurate, funded men and funded women should perform similarly after financing.
- **Taste-based discrimination:** investors have a preference against working with female founders, so women face a higher bar regardless of project quality.
- **Biased-belief stereotypes:** investors incorrectly evaluate founders based on sector-specific gender representations. Women are penalized in male-dominated sectors because they are less representative of the perceived successful founder type.

5.2 Empirical predictions

- If observable founder and startup characteristics explain the funding gap, adding those controls should eliminate the gender coefficient.
- If stereotypes are context-dependent, the gap should be stronger in male-dominated sectors and weaker or absent in female-dominated sectors.
- If statistical discrimination is the dominant mechanism, individual quality signals such as elite education or serial entrepreneurship should narrow the gender gap.
- If stereotypes are important, women who pass the funding screen should outperform men because investors set a higher bar for them.

- If equity investors but not banks exhibit gender-based biases, the gap should appear in VC/external equity but not in bank debt.

6 Empirical specifications

6.1 Baseline gender-gap regression

The first specification estimates gender differences in the use of each financing source:

$$\text{Funded}_i = \lambda_z + \lambda_{kt} + \delta \text{Female}_i + \beta' X_i + \varepsilon_i. \quad (1)$$

Here, Funded_i is an indicator for using a financing source, λ_z are county fixed effects, λ_{kt} are four-digit sector by cohort-year fixed effects, and X_i includes founder characteristics, startup characteristics, motivations, startup capital buckets, and business model controls.

6.2 Sector gender-congruence regression

The second specification interacts female founder status with sector gender composition:

$$\text{Funded}_i = \beta_1 \text{Female}_i + \beta_2 (\text{Female}_i \times \text{FemaleDominatedSector}_{kt}) + \delta_z + \delta_{kt} + \gamma' X_i + \varepsilon_i. \quad (2)$$

The coefficient β_1 captures the gender gap in male-dominated sectors. The sum $\beta_1 + \beta_2$ captures the gender gap in female-dominated sectors.

6.3 Oaxaca–Blinder decomposition

The paper decomposes the mean gender gap in external equity into explained and unexplained components:

$$\bar{Y}_m - \bar{Y}_f = \underbrace{\hat{\beta}_m(\bar{X}_m - \bar{X}_f)}_{\text{explained by characteristics}} + \underbrace{\bar{X}_f(\hat{\beta}_m - \hat{\beta}_f)}_{\text{unexplained component}}. \quad (3)$$

The unexplained component is interpreted as the part of the financing gap not accounted for by observable differences in founder and startup characteristics.

6.4 Outcome test

The performance specification compares future performance conditional on funding:

$$\text{FuturePerformance}_{i,t+\Delta} = \beta_1 \text{Female}_i + \beta_2 \text{Funded}_i + \beta_3 (\text{Female}_i \times \text{Funded}_i) + \delta_z + \delta_{kt} + \gamma' X_i + \varepsilon_{i,t}. \quad (4)$$

If investors set the same correct bar for male and female founders, funded male- and female-founded firms should perform similarly. If female founders face a higher bar, funded female-founded startups should outperform.

7 Main results and interpretation

7.1 Average gender gap in equity financing

Female founders are less likely to use external equity and VC even after rich controls. In the baseline specification, women are 0.56 percentage points less likely to use external equity relative to a mean of 2.54%, corresponding to a 22% relative gap. For VC, the coefficient is -0.11 percentage points relative to a mean of 0.30%, corresponding to a 36% relative gap.

Crucially, the same regression shows no negative gender gap in corporate bank loans or public grants. This contrast matters because it suggests that the equity gap is not simply driven by female founders having lower quality projects or less demand for external finance in general.

7.2 Sector heterogeneity

The gender gap is concentrated in male-dominated sectors. Female-founded startups in male-dominated sectors are 0.69 percentage points less likely to use external equity and 0.15 percentage points less likely to use VC. In female-dominated sectors, the interaction term offsets the baseline gap, so the equity gap is close to zero.

This pattern is consistent with context-dependent stereotypes. If the problem were only that female founders have lower growth ambitions on average, the gap should not disappear when women operate in female-dominated sectors after controlling for motivations and startup type.

7.3 Observable characteristics and unexplained gap

Oaxaca–Blinder decomposition shows that observable differences explain only part of the gap. Overall, 42% of the average external-equity gender gap is explained by differences in characteristics. In male-dominated sectors, 80% of the gap remains unexplained. In female-dominated sectors, the unexplained component is not statistically different from zero.

This is one of the paper’s most important findings. It shows that the gender gap is not merely about different founder characteristics, but about how those characteristics translate into funding outcomes in different sector contexts.

7.4 Quality signals do not close the gap

The paper further tests whether standard quality signals, such as serial entrepreneurship, incorporation, innovation, and growth orientation, narrow the gap for women. In male-dominated sectors, many of these signals have weaker or even negative interactions with female founder status. For example, serial female entrepreneurs are less likely to use external equity than male serial entrepreneurs.

Under pure statistical discrimination, individual signals of quality should reduce uncertainty and narrow the gap. The fact that they do not do so is more consistent with stereotypes or biased evaluation.

7.5 Substitution from equity to debt

Growth-oriented female founders in male-dominated sectors are more likely to use corporate bank loans. Among startups with growth potential, women using bank loans are less likely to use external equity, suggesting that bank loans substitute for equity. Male founders with similar profiles appear to use debt and equity more complementarily.

This result does not mean bank debt is an equal replacement for VC. Bank loans require repayment and collateral, while VC provides risk capital, advice, and networks. The substitution may therefore leave some high-risk, high-return female-led projects underfunded.

7.6 Outcome tests

VC-backed female-founded startups in male-dominated sectors outperform VC-backed male-founded startups. They are more likely to survive after three years and have higher three-year sales growth. This performance gap is not present for bank-financed startups, nor is it present in female-dominated sectors.

The outcome test supports the interpretation that VC investors set a higher bar for women in male-dominated sectors. If the bar were equal and based on true quality, funded male and female founders should perform similarly.

8 Tables and figures from the paper

8.1 Table 1: Male- and female-founded startups by cohort and sectors

How to read it. Table 1 documents the basic sector composition. Women found about 29% of firms, but their presence varies sharply across sectors. Female-dominated sectors account for roughly one-fifth of firms, while male-dominated sectors account for four-fifths. **Economic message:** The raw setting already contains gender segregation across entrepreneurial sectors. This sectoral sorting is essential because the paper’s stereotype test relies on whether the financing gap differs across male- and female-dominated sectors.

Table 1
Male- and female-founded startups by cohort and sectors
Panel A. Male and female entrepreneurs

	Cohorts					Total
	2002	2006	2010	2014	2018	
Male entrepreneurs	15,300	22,030	24,351	21,013	10,607	93,301
%	72.12	71.37	71.67	69.72	70.31	71.07
Female entrepreneurs	5,921	8,857	9,621	9,131	4,479	38,009
%	27.88	28.63	28.33	30.28	29.69	28.93
Total	21,216	30,867	33,975	30,140	15,086	131,284

Panel B. Entrepreneurs by male- and female-dominated sectors

	Cohorts					Total
	2002	2006	2010	2014	2018	
<u>Male-dominated sectors</u>	17,657	24,827	28,528	23,148	11,029	105,189
%	83.22	80.43	83.97	76.80	73.11	80.12
Male in M-dominated sectors	13,881	19,474	22,043	18,238	8,661	82,297
%	65.43	63.09	64.88	60.51	57.41	62.69
Female in M-dominated sectors	3,776	5,353	6,485	4,910	2,368	22,892
%	17.80	17.34	19.09	16.29	15.70	17.44
<u>Female-dominated sectors</u>	3,559	6,040	5,447	6,992	4,057	26,095
%	16.78	19.57	16.03	23.20	26.89	19.88
Male in F-dominated sectors	1,419	2,556	2,308	2,775	1,946	11,004
%	6.69	8.28	6.79	9.21	12.90	8.38
Female in F-dominated sectors	2,140	3,484	3,139	4,217	2,111	15,091
%	10.09	11.29	9.24	13.99	13.99	11.49
Total	21,216	30,867	33,975	30,140	15,086	131,284

Panel C. Ranking of one-digit French SIC by percentage of female entrepreneurs

Rank	Sector (one-digit French SIC)	# Startups	% Female-founded	% External equity
1	Human health and social work activities	7,586	0.628	0.009
2	Other service activities	8,283	0.606	0.016
3	Education	3,918	0.411	0.019
4	Arts, entertainment, and recreation	3,743	0.363	0.027
5	Accommodation and food service activities	10,280	0.342	0.031
6	Professional, scientific, and technical activities	13,719	0.329	0.022
7	Real estate activities	5,616	0.317	0.031
8	Wholesale and retail trade, repair of motor vehicles and motorcycles	23,876	0.312	0.025
9	Administrative and support service activities	9,115	0.276	0.025
10	Manufacturing	9,927	0.259	0.031
11	Agriculture, forestry, and fishing	1,309	0.259	0.011
12	Financial and insurance activities	2,220	0.247	0.036
13	Information and communication	6,017	0.174	0.035
14	Transportation and storage	5,192	0.168	0.027
15	Mining and quarrying	96	0.156	0.052
16	Water supply, sewerage, waste management, and remediation activities	643	0.148	0.047
17	Electricity, gas, steam, and air conditioning supply	1,298	0.145	0.064
18	Construction	25,423	0.08	0.024

Source: SINE survey and firm registry. Panel A reports the number and percentage of male- and female-founded startups. Panel B reports the distributions of male- and female-founded startups in male- and female-dominated sectors. A female-dominated sector is a four-digit French SIC that includes at least 50% of

Table 1: Male- and female-founded startups by cohort and sectors

8.2 Table 2: Entrepreneur information and startup characteristics

How to read it. Table 2 compares male and female founders across education, experience, motivations, business models, funding sources, startup capital, and outcomes. Women are more educated on average but less likely to be elite-school graduates, serial entrepreneurs, industry experts, growth-oriented, incorporated, B2B-focused, and equity-backed. **Economic message:**

The table motivates the need for rich controls. It also shows why raw gaps alone are not enough: men and women differ on observable dimensions that investors may value, but these differences still do not fully explain the equity gap.

Variables	All	Male		Female		Difference	T-stat
	Mean	N	Mean	N	Mean		
<i>Panel A: Biographical characteristics</i>							
Age ≥ 40	0.466	93,301	0.481	37,983	0.431	0.05****	16.54
French	0.906	93,301	0.897	37,983	0.926	-0.03****	-17.19
<i>Education:</i>							
No degree	0.190	93,301	0.209	37,983	0.143	0.07****	29.35
High school	0.410	93,301	0.429	37,983	0.363	0.07****	22.11
Undergraduate	0.153	93,301	0.135	37,983	0.198	-0.06****	-27.13
Graduate	0.248	93,301	0.228	37,983	0.296	-0.07****	-24.94
Elite school	0.049	93,301	0.056	37,983	0.029	0.03****	24.10
<i>Experience:</i>							
Industry expert	0.601	93,301	0.635	37,983	0.518	0.12****	38.68
Serial entrepreneur	0.300	93,301	0.332	37,983	0.221	0.11****	41.95
Industry expert (≥ 10 years)	0.347	93,301	0.381	37,983	0.262	0.12****	42.99
Serial entrepreneur (≥ 3 startups)	0.032	93,301	0.040	37,983	0.014	0.03****	28.97
<i>Family:</i>							
Children	0.555	53,650	0.549	22,443	0.570	-0.02****	-5.30
Married	0.706	78,001	0.720	32,067	0.673	0.05****	15.31
Cofounder(s)	0.245	93,301	0.248	37,983	0.238	0.01****	3.75
Cofounded with spouse	0.093	93,301	0.090	37,983	0.101	-0.01****	-6.18
Cofounded with family	0.040	93,301	0.040	37,983	0.040	-0.00	-0.08
Cofounded with business partners	0.120	93,301	0.126	37,983	0.106	0.02****	10.09
<i>Panel B: Motivations and growth expectations</i>							
Growth-oriented	0.349	93,301	0.377	37,983	0.281	0.10****	34.21
Optimistic entrepreneur	0.158	56,908	0.178	24,434	0.112	0.07****	25.33
Incorporated	0.531	93,301	0.566	37,983	0.445	0.12****	39.79
<i>Detailed motivations for entry:</i>							
Independence	0.626	93,301	0.629	37,983	0.620	0.01****	3.05
Taste	0.437	93,301	0.450	37,983	0.404	0.05****	15.28
Opportunity	0.196	93,301	0.190	37,983	0.211	-0.02****	-8.78
New idea	0.162	93,301	0.161	37,983	0.165	-0.00	-1.41
Successful peer	0.102	93,301	0.104	37,983	0.097	0.01****	3.68
Unemployed	0.224	93,301	0.214	37,983	0.249	-0.04****	-13.57
Other reasons	0.300	93,301	0.296	37,983	0.308	-0.01****	-4.30
<i>Panel C: Startup characteristics and business model</i>							
<i>Innovation:</i>							
Innovative business	0.433	93,301	0.426	37,983	0.452	-0.03****	-8.66
Innovative product	0.347	93,301	0.336	37,983	0.375	-0.04****	-13.33
Innovative production	0.097	93,301	0.105	37,983	0.080	0.02****	14.52
Innovative marketing	0.133	78,001	0.133	32,067	0.131	0.00	0.77
Innovative organization	0.154	60,664	0.160	24,667	0.139	0.02****	7.99
Innovative sector	0.046	93,291	0.055	38,009	0.025	0.03****	27.70
<i>Business model:</i>							
B2B business model	0.388	93,301	0.430	37,983	0.286	0.14****	51.10
Nonlocal clientele	0.438	93,301	0.477	37,983	0.341	0.13****	45.03
Domestic customers	0.381	93,301	0.418	37,983	0.290	0.13****	45.03
International customers	0.057	93,301	0.059	37,983	0.051	0.01****	6.07
Local customers	0.562	93,301	0.523	37,983	0.659	-0.14****	-46.37
1 or 2 customers	0.141	93,298	0.152	37,980	0.113	0.04****	19.46
3 to 10 customers	0.270	93,298	0.284	37,980	0.235	0.05****	18.70
Many customers	0.455	93,298	0.425	37,980	0.528	-0.10****	-34.18
Many customers, a few large ones	0.135	93,298	0.140	37,980	0.124	0.02****	7.56
<i>Panel D: Startup funding and startup capital</i>							
<i>External funding sources:</i>							
External financing	0.493	93,301	0.493	37,983	0.495	-0.00	-0.76
External equity	0.025	93,301	0.029	37,983	0.017	0.01****	14.22
VC	0.003	55,971	0.004	23,230	0.001	0.00****	5.93
Business equity	0.022	93,301	0.025	37,983	0.015	0.01****	12.70
Bank loans	0.271	93,301	0.276	37,983	0.259	0.02****	6.31
Household debt	0.121	93,301	0.117	37,983	0.128	-0.01****	-5.55
Subsidized loans	0.082	93,301	0.079	37,983	0.089	-0.01****	-5.90
Public grants	0.204	93,301	0.200	37,983	0.214	-0.01****	-5.32
Microcredit & crowdfunding	0.017	55,971	0.015	23,230	0.021	-0.01****	-5.38

(a) Table 2, first part

Variables	All	Male		Female		Difference	T-stat
	Mean	N	Mean	N	Mean		
<i>% Funding sources:</i>							
% Personal resources	62.276	24,340	62.790	9,621	60.976	1.81***	3.69
% Bank loans	30.391	24,340	29.946	9,621	31.514	-1.57***	-3.46
% Other financing	7.333	24,340	7.264	9,621	7.510	-0.25	-1.07
<i>Startup capital:</i>							
Startup capital < 2k	0.241	93,301	0.222	37,983	0.288	-0.07***	-24.64
Startup capital 2-8k	0.290	93,301	0.293	37,983	0.281	0.01***	4.69
Startup capital 8-16k	0.169	93,301	0.176	37,983	0.151	0.03***	11.47
Startup capital 16-40k	0.156	93,301	0.161	37,983	0.145	0.02***	7.28
Startup capital 40-80k	0.066	93,301	0.066	37,983	0.065	0.00	0.80
Startup capital 80-160k	0.043	93,301	0.043	37,983	0.042	0.00	0.66
Startup capital ≥160k	0.036	93,301	0.039	37,983	0.029	0.01***	9.59
<i>Other sources of income:</i>							
Spouse income	0.314	52,595	0.306	20,975	0.337	-0.03***	-8.13
Other employment income	0.160	52,595	0.161	20,975	0.158	0.00	1.03
Other income (benefits)	0.116	52,595	0.119	20,975	0.109	0.01***	3.85
No other income	0.482	52,595	0.488	20,975	0.467	0.02***	5.29
<i>Panel E: Startup performance</i>							
Survival ≥3 years	0.658	93,301	0.663	37,983	0.646	0.02***	6.13
Survival ≥5 years	0.380	93,301	0.385	37,983	0.369	0.02***	5.25
<i>Employment size:</i>							
Zero	0.807	93,115	0.795	37,894	0.838	-0.04***	-18.58
1	0.093	93,115	0.098	37,894	0.082	0.02***	9.19
2	0.042	93,115	0.045	37,894	0.035	0.01***	7.98
3	0.020	93,115	0.022	37,894	0.017	0.00***	5.43
4	0.020	93,115	0.022	37,894	0.017		
4-5	0.019	93,115	0.020	37,894	0.014	0.01***	8.46
6-10	0.013	93,115	0.014	37,894	0.009	0.01***	7.97
11+	0.006	93,115	0.007	37,894	0.005	0.00***	4.60
$\Delta sales_{t=0,3}$	0.5496	48,209	0.5627	19,168	0.5169	0.05***	6.01
$\Delta sales_{t=0,5}$	0.5383	26,637	0.5543	10,185	0.4964	0.06***	4.78
$\Delta employees_{t=0,3}$	0.1758	11,978	0.1883	3,661	0.1349	0.05***	4.57
$\Delta employees_{t=0,5}$	0.3419	6,158	0.3634	1,827	0.2696	0.09***	5.15

Source: SINE survey and tax files. Sample: New firms founded in 2002, 2006, 2010, 2014, and 2018. The table presents entrepreneurs' biographical characteristics (panel A), startup characteristics and business model information (panel B), entrepreneurs' motivations and growth preferences (panel C), startups' funding sources and initial capital (panel D), and startup performance (panel E). The mean and number of observations by gender mean are reported as one t-statistic and a value of the mean difference between male and female

(b) Table 2, continued

Table 2: Entrepreneur information and startup characteristics

8.3 Table 3: Gender gap and entrepreneur financing within sector

How to read it. Table 3 estimates the baseline gender gap within sector, cohort, and county, controlling for founder and startup characteristics. The dependent variables are external equity, VC, any external financing, corporate bank loans, personal debt, subsidized loans, and grants.

Economic message: The negative female coefficient appears only for equity and VC, not for bank loans or grants. This is the first piece of evidence that the gender gap is specific to equity

investors rather than to all capital providers.

Table 3
Gender gap and entrepreneur financing within sector

Dependent variable:	External equity		All external	Bank loans		Subsidized financing	
	All (1)	VC only (2)	Financing (3)	Corporate debt (4)	Personal debt (5)	Loans (6)	Grants (7)
<i>Female</i>	-0.0056*** (0.001)	-0.0011*** (0.000)	0.0049 (0.005)	0.0061 (0.004)	0.0053** (0.002)	0.0132*** (0.002)	0.0054 (0.003)
<i>French</i>	0.0003 (0.002)	0.0004 (0.001)	0.0870*** (0.005)	0.0735*** (0.005)	0.0170*** (0.003)	0.0022 (0.002)	0.0511*** (0.003)
<i>Age ≥ 40</i>	0.0008 (0.001)	0.0001 (0.000)	-0.0333*** (0.004)	-0.0366*** (0.002)	-0.0029 (0.002)	-0.0100*** (0.002)	-0.0095*** (0.003)
<i>Undergraduate</i>	0.0012 (0.001)	-0.0022*** (0.000)	-0.0001 (0.004)	0.0016 (0.005)	-0.0051* (0.003)	0.0021 (0.003)	0.0176*** (0.004)
<i>Graduate</i>	0.0051*** (0.002)	0.0008 (0.001)	-0.0290*** (0.005)	-0.0319*** (0.003)	-0.0142*** (0.003)	-0.0024 (0.002)	0.0133*** (0.003)
<i>Elite school</i>	0.0106*** (0.004)	0.0081*** (0.002)	-0.0356*** (0.007)	-0.0419*** (0.006)	-0.0195*** (0.004)	-0.0046 (0.004)	0.0034 (0.005)
<i>Industry expert</i>	0.0017* (0.001)	-0.0010** (0.000)	0.0209*** (0.004)	0.0306*** (0.003)	0.0044** (0.002)	0.0058*** (0.002)	-0.0103*** (0.003)
<i>Serial entrepreneur</i>	0.0090*** (0.001)	0.0010** (0.000)	-0.0539*** (0.004)	-0.0370*** (0.004)	-0.0034* (0.002)	-0.0193*** (0.002)	-0.0619*** (0.003)
<i>Cofounded</i>	0.0131*** (0.001)	0.0026*** (0.001)	-0.0188*** (0.003)	0.0068* (0.004)	-0.0153*** (0.003)	-0.0065*** (0.002)	-0.0280*** (0.003)
<i>Incorporated</i>	0.0135*** (0.001)	0.0002 (0.000)	-0.0352*** (0.004)	0.0529*** (0.006)	-0.0590*** (0.005)	-0.0028 (0.002)	-0.0523*** (0.003)
<i>Innovative business</i>	0.0030*** (0.001)	0.0016*** (0.000)	0.0467*** (0.003)	0.0123*** (0.003)	0.0173*** (0.002)	0.0139*** (0.002)	0.0476*** (0.002)
<i>B2B business model</i>	0.0054*** (0.001)	-0.0008 (0.001)	-0.0257*** (0.004)	-0.0158*** (0.004)	-0.0127*** (0.002)	-0.0018 (0.002)	-0.0084*** (0.003)
<i>Nonlocal clientele</i>	0.0007 (0.001)	0.0010** (0.000)	-0.0462*** (0.004)	-0.0407*** (0.003)	-0.0076*** (0.002)	-0.0084*** (0.002)	-0.0158*** (0.003)
<i>Growth-oriented</i>	0.0123*** (0.001)	0.0018*** (0.000)	0.0064** (0.003)	-0.0020 (0.003)	0.0021 (0.002)	0.0058*** (0.002)	-0.0055** (0.002)
<i>Motivation independence</i>	-0.0070*** (0.001)	0.0002 (0.000)	0.0329*** (0.003)	0.0243*** (0.003)	0.0079*** (0.002)	0.0150*** (0.002)	0.0243*** (0.002)
<i>Motivation taste</i>	0.0002 (0.001)	-0.0004 (0.000)	0.0282*** (0.003)	0.0249*** (0.003)	0.0104*** (0.003)	0.0184*** (0.002)	0.0135*** (0.002)
<i>Motivation opportunity</i>	0.0045*** (0.001)	-0.0005 (0.001)	0.0161*** (0.004)	0.0106*** (0.003)	0.0014 (0.002)	0.0067*** (0.002)	0.0066** (0.003)
<i>Motivation new idea</i>	0.0079*** (0.002)	0.0035*** (0.001)	0.0214*** (0.004)	0.0131*** (0.004)	-0.0020 (0.003)	0.0252*** (0.003)	0.0289*** (0.003)
<i>Motivation successful peers</i>	0.0018 (0.002)	0.0018** (0.001)	0.0173*** (0.004)	0.0117*** (0.004)	0.0005 (0.003)	0.0080*** (0.003)	0.0045 (0.003)
<i>Motivation unemployed</i>	-0.0049*** (0.001)	0.0011** (0.001)	0.0932*** (0.005)	0.0010 (0.003)	0.0116*** (0.002)	0.0417*** (0.003)	0.1434*** (0.005)

Source: SINE survey and firm registry. Sample: New firms founded in 2002, 2006, 2010, 2014, and 2018. This table uses OLS to analyze the gender gap in entrepreneurial finance. The dependent variables are as follows: *External equity* financing including VC and other sources of business equity (column 1), *VC only* (column 2), *External financing* of any kind (column 3), *Bank corporate loans* (column 4), *Personal bank debt* (column 5), *Subsidized loans* (column 6), and *Public grants* (column 7). *Female* is a dummy variable equal to one if a woman runs the startup. The baseline controls include the following dummy variables: *Undergraduate*, *Graduate*, *Elite school*, and *Industry expert* equal one if the entrepreneur has at least 3 years of work experience in the sector; *Serial entrepreneur* equals one if the entrepreneur has previously founded another startup; *Incorporated startup* equals one if the startup is incorporated instead of a sole proprietorship; *Growth-oriented* equals one if the entrepreneur's main ambition is to "develop the startup" as opposed to "create their own job"; *Innovative business* equals one if the entrepreneur reports any innovation in his business model; *B2B* equals one if the startup is a B2B firm as opposed to B2C firm. *Nonlocal clients* equals one if the startup's customer base is national or foreign as opposed to local. The models also include motivation items that stem from the survey question "What are your three main motivations to start a business?". All models include county and four-digit SIC sector × cohort-year fixed effects and startup capital dummies. Clustered standard errors at the sector level are reported in parentheses.

Table 3: Gender gap and entrepreneur financing within sector

8.4 Table 4: Gender stereotypes and entrepreneur financing

How to read it. Table 4 interacts female founder status with female-dominated sector status. Panel A presents the baseline model. Panel B adds hard balance-sheet information, including total assets and tangibility. **Economic message:** The female coefficient is negative in male-dominated sectors and the interaction with female-dominated sectors offsets the gap. Adding hard information does not eliminate the equity gap, while tangibility strongly predicts bank loans. This supports the idea that banks and VCs rely on different information.

Table 4
Gender stereotypes and entrepreneur financing

Panel A: Baseline model

Dependent variable:	External equity		All external	Bank loans			Subsidized financing	
	All (1)	VC only (2)		Financing (3)	Corporate debt (4)	Personal debt (5)	Loans (6)	Grants (7)
<i>Female</i>	-0.0069*** (0.001)	-0.0015*** (0.000)	0.0048 (0.005)	0.0024 (0.004)	0.0042* (0.002)	0.0124*** (0.002)	0.0005 (0.003)	0.0024 (0.003)
<i>Female × F-dominated sector</i>	0.0059*** (0.002)	0.001*** (0.001)	0.0269*** (0.009)	0.0144* (0.008)	0.0042 (0.004)	0.0022 (0.005)	0.0187*** (0.007)	0.0005 (0.003)
Founders controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Startup characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Startup capital	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Motivations	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0497	0.0329	0.2727	0.2425	0.0908	0.0652	0.2534	0.2534
N	131,284	79,291	131,284	131,284	131,284	131,284	131,284	131,284
Mean dep. var.	0.0254	0.0030	0.4933	0.2711	0.1205	0.0819	0.2043	0.2043

(a) Panel A: Baseline model

Table 4
(Continued)

Panel B: Role of hard information

Dependent variable:	External equity		All external	Bank loans		Subsidized financing	
	All (1)	VC only (2)		Financing (3)	Corporate debt (4)	Personal debt (5)	Loans (6)
<i>Female</i>	-0.0086*** (0.002)	-0.0027*** (0.001)	0.0087 (0.008)	0.0012 (0.005)	0.0006*** (0.000)	0.0153*** (0.003)	0.0024 (0.004)
<i>Female × F-dominated sector</i>	0.0093*** (0.004)	0.0026* (0.001)	0.0301*** (0.009)	0.0202** (0.009)	0.0061 (0.007)	0.0067 (0.006)	0.0132 (0.008)
<i>Intangible assets</i>	0.0015*** (0.000)	-0.0002 (0.000)	-0.0002 (0.001)	0.0009*** (0.001)	-0.0010 (0.001)	-0.0011* (0.001)	-0.0044*** (0.001)
<i>Tangible/intangible assets</i>	-0.0035 (0.003)	-0.0027* (0.002)	0.0796*** (0.008)	0.0916*** (0.008)	0.0148** (0.007)	0.0166*** (0.006)	0.0491*** (0.007)
Founder and Startup characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0500	0.0334	0.2708	0.2305	0.0842	0.0666	0.2738
N	81,610	57,102	81,610	81,610	81,610	81,610	81,610
Mean dep. var.	0.0275	0.0032	0.5025	0.3005	0.1246	0.0877	0.1794

Source: SINE survey and firm registry. Sample: New firms founded in 2002, 2006, 2010, 2014, and 2018. This table uses OLS to analyze the effect of gender stereotypes on the use of funding sources available to startups. The dependent variables are as follows: *External equity* financing including VC (column 1), VC only (column 2), *External financing* of any kind (column 3), *Bank corporate loans* (column 4), *Personal bank debt* (column 5), *Subsidized loans* (column 6), and *Public grants* (column 7). *Female* is a dummy variable equal to one if a woman runs the startup. *F-dominated sector* is a dummy variable equal to one if the sector includes at least 50% of startups within a four-digit SIC sector that women found. All models include business controls as in Table 1 and county and four-digit SIC sector × cohort-year fixed effects. Specifically, Founders' controls include age, citizenship, education, and experience dummies. Startup characteristics include the incorporation status, innovative business, cofounders, R&D, and technical climate dummies. Startup capital includes buckets of the initial capital amount received by the entrepreneur, and motivations include growth-orientation and its motivation dummies. In addition, panel B includes the logarithm of total assets and the ratio of tangible assets. Clustered standard errors in this column based are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

(b) Panel B: Role of hard information

Table 4: Gender stereotypes and entrepreneur financing

8.5 Table 5: Gender stereotypes and external equity financing - thresholds of female-dominated sectors

How to read it. Table 5 checks whether the equity gap varies continuously with the percentage of women in the sector and whether the gap is especially strong in innovative sectors. **Economic**

message: The gap narrows as the share of women in the sector rises, and it is especially severe in innovative sectors, which are heavily male-dominated. This strengthens the context-dependent stereotype interpretation.

8.6 Table 6: Gender stereotypes and omitted variables

How to read it. Table 6 implements robustness checks for omitted variables using coefficient stability and Oster-style bounds. **Economic message:** The coefficient on female founder sta-

tus remains economically meaningful and statistically significant after adding controls. Omitted variables would need to be very influential to fully explain away the external-equity and VC gaps.

	Large Bins (1)	Small Bins (2)	Continuous (3)	Innovative Sectors (4)
<i>Female</i>	-0.0085*** (0.002)	-0.0085*** (0.002)	-0.0093*** (0.002)	-0.0047*** (0.001)
<i>Female</i> × Sector [20–40]% Female-founded	0.0045 (0.003)	0.0045 (0.003)		
<i>Female</i> × Sector [40–60]% Female-founded	0.0021 (0.003)			
<i>Female</i> × Sector [60–80]% Female-founded	0.0082** (0.004)			
<i>Female</i> × Sector [40–50]% Female-founded		-0.0013 (0.003)		
<i>Female</i> × Sector [50–60]% Female-founded		0.0079** (0.004)		
<i>Female</i> × Sector [60–70]% Female-founded		0.0090* (0.005)		
<i>Female</i> × Sector [70–80]% Female-founded		0.0072* (0.004)		
<i>Female</i> × Sector > 80% Female-founded	0.0030 (0.004)	0.0030 (0.004)		
<i>Sector's % Female-founded firms</i>			-0.0052 (0.003)	
<i>Female</i> × Sector's % Female-founded			0.0099*** (0.003)	
<i>Female</i> × Innovative sector				-0.0182*** (0.004)
Sector × Cohort-year FE	Yes	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
R^2	0.0512	0.0512	0.0293	0.0512
N	131,258	131,258	131,258	131,258
Mean dep. var.	0.0253	0.0253	0.0253	0.0253

Source: SINE survey and firm registry. *Sample:* New firms founded in 2002, 2006, 2010, 2014, and 2018. This table uses OLS to test different thresholds of female-dominated sectors on the use of external equity. The dependent variable is *External equity* financing including VC and other sources of business equity. *Female* is a dummy variable equal to one if a woman runs the startup. It is interacted with dummy variables associated with sectors categorized with their % of women-founded firms (columns 1 and

(a) Table 5: Thresholds and innovative sectors

Table 6
Gender stereotypes and omitted variables

Panel A: Effects of controls

Dependent variable:	<i>I(External equity)</i>			<i>I(VC)</i>		
	Unrestricted	Restricted		Unrestricted	Restricted	
Model:	None (1)	Fixed effects (2)	Baseline (3)	None (4)	Fixed effects (5)	Baseline (6)
Controls:						
<i>Female</i>	-0.0122*** (0.001)	-0.0097*** (0.001)	-0.0069*** (0.001)	-0.0021*** (0.000)	-0.0017*** (0.000)	-0.0015*** (0.000)
Human capital	No	No	Yes	No	No	Yes
Startup characteristics	No	No	Yes	No	No	Yes
Business model	No	No	Yes	No	No	Yes
Startup capital	No	No	Yes	No	No	Yes
Motivations	No	No	Yes	No	No	Yes
Sector × Cohort-year FE	No	Yes	Yes	No	Yes	Yes
County FE	No	Yes	Yes	No	Yes	Yes
R^2	0.0012	0.0259	0.0497	0.0003	0.0227	0.0329
N	131,284	131,284	131,284	79,201	79,201	79,201
Mean dep. var.	0.0254	0.0254	0.0254	0.0030	0.0030	0.0030

Panel B: Oster's (2019) test for omitted variables in the External Equity model

Parameters:	$R_{max}^2 = \min(2 \times R_{\mu}^2, 1) = 0.0994$ and $\delta = 1$							
Model:	Uncontrolled effect		Controlled effect		Identified set	Reject	δ_{max} s.t. $\beta=0$	
Controls:	None		Augmented		[bound1 ; bound2]	null?		
Treatment variable:	β_u	R_{μ}^2	β_c	R_c^2	[β_c ; β_{adj}]			
Female	-0.0122	0.0012	-0.0069	0.0497	-0.0069	-0.0015	Yes	1.270

Panel C: Oster's (2019) test for omitted variables in the VC model

Choice of parameters:	$R_{max}^2 = \min(2 \times R_c^2, 1) = 0.0658$ and $\delta = 1$							
Model:	Uncontrolled effect		Controlled effect		Identified set		Reject null?	$\hat{\alpha}_{max}$ s.t. $\beta=0$
Controls:	None		Augmented		[bound1 ; bound2]			
Treatment variable:	β_u	R_u^2	β_c	R_c^2	[$\beta_c ; \beta_{adj}$]			
Female	-0.0021	0.0003	-0.0015	0.0329	-0.0015	-0.0009	Yes	2.477

Source: SINE survey and firm registry. *Sample:* New firms founded in 2002, 2006, 2010, 2014, and 2018. This table tests the role of omitted variables in the equity gender gap. Panel A uses OLS to test the robustness of the model, including control variables. I estimate the linear models $External\ equity = \beta X + \epsilon$ and $VC = \beta X + \epsilon$ without any control variables to obtain β_u and R_u^2 (columns 1 and 4), then with the baseline fixed effects (columns 2 and 5), and with control variables of Tables 3 and 4 (columns 3 and 6). Panel B implements the test designed by Oster (2019) to assess the robustness of the baseline model (panel A, columns 3 and 6) to the omitted variable bias. For any given combination δ and R_{max}^2 , Oster (2019) defines the bias-adjusted coefficient β_{adj} that is given by the following equation: $\beta_{adj} = \beta_c - \delta \frac{(\beta_u - \beta_c) \times (R_{max}^2 - R_c^2)}{R_c^2 - R_u^2}$. Assumptions of the model are $\delta = 1$, which means that

(b) Table 6: Omitted-variable bounds

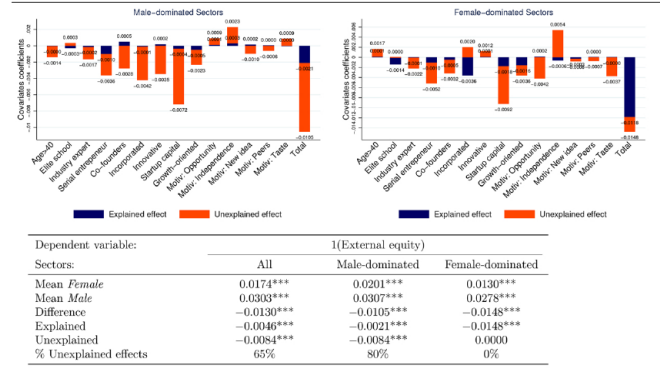
Tables 5 and 6: Robustness for sector heterogeneity and omitted variables

8.7 Table 7: Quantifying gender stereotypes across male- and female-dominated sectors

How to read it. Table 7 reports the Oaxaca–Blinder decomposition of the external-equity gap into explained and unexplained components. It separately plots results for male- and female-dominated sectors. **Economic message:** In male-dominated sectors, the unexplained component dominates.

In female-dominated sectors, observable characteristics explain the gap. This table is the paper's cleanest quantitative summary of the stereotype interpretation.

Table 7
Quantifying gender stereotypes across male- and female-dominated sectors



Source: SINE survey and firm registry. *Sample:* New firms founded in 2002, 2006, 2010, 2014, and 2018. The figure shows the results of the Blinder–Oaxaca decomposition of the difference in the use of external equity and VC financing between male and female entrepreneurs in male- and female-dominated sectors. The Oaxaca–Blinder decomposition decomposes the mean difference in the use of external equity between male- and female-founded startups for two components: the “explained effect” (dark blue) and the “unexplained effect” (orange). They are estimated using separate OLS regressions for male and female entrepreneurs. The explained effects correspond to the effect of gender differences in the explanatory variables evaluated using the male equation coefficients. The unexplained effects correspond to the average female residuals from the male equation. The top figure corresponds to Oaxaca–Blinder decomposition in male-dominated sectors, and the bottom figure corresponds to Oaxaca–Blinder decomposition in female-dominated sectors. In addition to the displayed variables, the estimated models include the following groups of variables: *Age and citizenship*; Education includes *Undergraduate, Graduate, and Elite school*; *Industry expert*; *Serial entrepreneur*; *Incorporation status*; Team composition includes *Cofounded* and *Founded with Spouse, Relatives, or Business partners*; Growth preferences include *growth-oriented* and *detailed ex-ante motivations*; Business model includes *B2B*, *Number of customers*, and *Non-local customers*; Startup capital includes the categories of *startup capital*; Other funding includes *alternative financing sources*; and *SIC-4* sectors. The coefficients are reported in [Internet Appendix Table IA16](#). [Internet Appendix Figure 1A11](#) plots the Blinder–Oaxaca decomposition results for all sectors without sector fixed effects. [Internet Appendix Figure 1A12](#) includes family variables. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 7: Quantifying gender stereotypes across male- and female-dominated sectors

8.8 Table 8: Gender stereotypes and financing highly skilled and motivated female entrepreneurs

How to read it. Table 8 asks whether high-quality founder signals mitigate the equity gap. Panel A focuses on external equity and VC-related characteristics. Panel B examines alternative financing sources. **Economic message:** Quality signals such as serial entrepreneurship, incorporation, innovation, and growth orientation do not close the equity gap for women in male-dominated sectors. In some cases, the interaction is negative. Bank loans, however, are used more by female founders with high-growth profiles, consistent with substitution.

(Continued)									
Panel B: Alternative financing sources									
Dependent variable:	I (Bank loans)		II (Personal debt)		III (Subsidized loans)		IV (Public grants)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
Female	-0.0025 (0.005)	-0.0018 (0.004)	0.0001 (0.002)	0.0005 (0.003)	0.0103*** (0.002)	0.0103*** (0.002)	-0.0006 (0.006)	-0.0012 (0.004)	
Female × Serial entrepreneur	0.0111** (0.008)	0.0126** (0.006)	0.0120** (0.006)	0.0120** (0.006)	0.0069 (0.004)	0.0069 (0.004)	0.0148** (0.006)	-0.0114** (0.005)	
High-growth, innovative, incorporated startup	-0.0231*** (0.006)	-0.0231*** (0.006)	-0.0068 (0.006)	-0.0068 (0.006)	-0.0103*** (0.004)	-0.0103*** (0.004)	-0.0114** (0.006)	-0.0114** (0.005)	
Female × IHH startup	0.0277*** (0.010)	0.0277*** (0.010)	-0.0096 (0.007)	-0.0096 (0.007)	0.0155** (0.007)	0.0155** (0.007)	0.0000 (0.008)	0.0000 (0.008)	
Serial entrepreneur	-0.0419*** (0.005)	-0.0383*** (0.004)	-0.0067*** (0.002)	-0.0067*** (0.002)	-0.0220*** (0.002)	-0.0220*** (0.002)	-0.0640*** (0.002)	-0.0612*** (0.002)	
R ²	0.2398	0.2399	0.0768	0.0768	0.0637	0.0637	0.2758	0.2758	
N	105,165	105,165	105,165	105,165	105,165	105,165	105,165	105,165	
Mean dep. var.	0.2741	0.2741	0.1181	0.1181	0.0831	0.0831	0.2063	0.2063	
Founder controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Startup characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Startup capital	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Motivations	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Sector × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Control FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	

Source: SINE survey and firm registry. *Sample:* New firms founded in male-dominated sectors in 2002, 2006, 2010, 2014, and 2018. This table uses OLS to test whether entrepreneur and startup characteristics mitigate the effects of gender stereotypes on the use of external equity (panel A) and other financing sources (panel B). The dependent variables are *External equity* (panel A), *Bank debt*, *Personal debt*, *Subsidized debt*, and *Public grants* (panel B). The independent variable *Female* is interacted with the following items that are dummy variables that capture entrepreneur skills and motivations: *Serial entrepreneur* (column 1), *Industry expert* (column 2), *Incorporated startup* (column 3), *Innovative business* (column 4), *Growth-oriented* (column 5), *Optimistic entrepreneur* (column 6), whether the entrepreneur *Has children* (column 7), and *Cofounded startup* (column 8). Panel B includes the dummy variable *High growth, innovative, and incorporated (IHH) startups*, which equals one if the startup is innovative, an incorporated business, and whose founder's main objective is to “develop the business” as opposed to “create their own job.” Coefficients are reported in [Internet Appendix Tables IA14](#). All models include baseline controls as in [Table 3](#) and control for *Year dummies*, *Year dummies*, *Year dummies*, and *Year dummies*.

(a) Panel A: External equity and VC

(b) Panel B: Alternative financing sources

Table 8: Gender stereotypes and financing highly skilled and motivated female entrepreneurs

8.9 Table 9: Substitution between external equity and other financing sources

How to read it. Table 9 directly tests whether alternative funding sources complement or substitute for external equity. It separates average firms from startups with growth potential. **Economic**

message: Among growth-potential startups, female founders using bank loans are less likely to use external equity, suggesting substitution from equity to debt. This is important because the paper does not simply show that women receive less VC; it shows how they adjust financing strategy when equity is less accessible.

Dependent variable	I(External equity)							
Sample:	Average Firms				Startups with Growth Potential			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female	-0.006*** (0.001)	-0.009*** (0.001)	-0.007*** (0.001)	-0.006*** (0.001)	-0.010*** (0.006)	-0.014*** (0.005)	-0.012*** (0.005)	-0.0179*** (0.005)
Bank loan	0.0059*** (0.002)				-0.0009 (0.005)			
Female × Bank loans	-0.0030 (0.003)				-0.0196** (0.009)			
Personal loan		-0.0024 (0.002)				0.0074 (0.008)		
Female × Personal loans		-0.0071** (0.003)				-0.0267*** (0.010)		
Subsidized loans			0.0020 (0.002)				-0.0072 (0.009)	
Subsidized loans × Female			0.0035 (0.005)				-0.0018 (0.017)	
Public grant				0.0020 (0.002)				0.0048 (0.009)
Female × Public grants				-0.0025 (0.002)				-0.0121 (0.012)
Founders controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Startup characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Startup capital	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Motivations	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.0493	0.0492	0.0492	0.0491	0.1121	0.1120	0.1119	0.1118
N	105,165	105,165	105,165	105,165	14,561	14,561	14,561	14,561
Mean dep. var.	0.0271	0.0271	0.0271	0.0271	0.0611	0.0611	0.0611	0.0611

Source: SINE survey and firm registry. Sample: New firms founded in male-dominated sectors in 2002, 2006, 2010, 2014, and 2018. This table uses OLS to test whether entrepreneurs use alternative funding sources to complement or substitute external equity financing. The sample in columns (1) to (4) excludes startups with growth potential. The sample in columns (5) to (8) includes only startups with growth potential, which are defined as founded by a growth-oriented entrepreneur who starts an innovative and incorporated startup. The dependent variable is External equity, which equals one if the startup uses VC or other sources of external equity, and zero otherwise. The independent variable Female is interacted with alternative funding sources: Bank loans (columns 1 and 5), Personal debt (columns 2 and 4), Subsidized loans (columns 3 and 6), and Public grants (columns 4 and 8). All models include the same baseline controls as in Table 3, and county, four-digit SIC sector × cohort-year fixed effects. Clustered standard errors at the four-digit SIC sector level.

Table 9: Substitution between external equity and other financing sources

8.10 Table 10: Outcome test of equity-backed and bank-financed startups

How to read it. Table 10 compares the post-financing performance of funded male- and female-founded startups in male- and female-dominated sectors. The key outcomes are survival and sales growth over three and five years. **Economic message:** VC-backed female-founded startups in

male-dominated sectors outperform male-founded startups in short-run survival and sales growth. This pattern is not found for bank-financed firms. The result suggests that VC investors set a higher funding threshold for women, while banks do not show the same pattern.

Panel A: Male-dominated sectors									
Financing	VC-backed startups				Bank-financed startups				
Dependent variable	I(Survival)		Δ sales		I(Survival)		Δ sales		
	≥3 (1)	≥5 (2)	(0,3) (3)	(0,5) (4)	≥3 (5)	≥5 (6)	(0,3) (7)	(0,5) (8)	
Female	-0.0172*** (0.01)	-0.0062 (0.009)	-0.0062*** (0.01)	-0.0780*** (0.02)	-0.0209*** (0.01)	-0.0155*** (0.009)	-0.0241*** (0.01)	-0.0449* (0.02)	
VC	-0.0199 (0.03)	0.0005 (0.02)	0.0003 (0.12)	0.4185** (0.17)					
Female × VC	0.1225** (0.05)	0.0400 (0.04)	0.6199** (0.28)	0.2261 (0.22)					
Bank loans					0.0124*** (0.01)	0.0094** (0.009)	0.0522*** (0.02)	0.0563*** (0.01)	
Female × Bank loans					-0.0068 (0.01)	-0.0090 (0.01)	-0.0270 (0.02)	-0.0399 (0.04)	
Startup & Founders characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Sector FE × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
R ²	0.365	0.450	0.077	0.109	0.281	0.354	0.074	0.108	
N	62,705	62,705	34,129	15,184	105,165	105,165	51,456	29,480	
Mean dep. var.	0.0004	0.2518	0.5270	0.3602	0.6618	0.3856	0.5552	0.5317	

(a) Panel A: Male-dominated sectors

Panel B: Female-dominated sectors									
Financing	VC-backed startups				Bank-financed startups				
Dependent variable	I(Survival)		Δ sales		I(Survival)		Δ sales		
	≥3 (1)	≥5 (2)	(0,3) (3)	(0,5) (4)	≥3 (5)	≥5 (6)	(0,3) (7)	(0,5) (8)	
Female	0.0055 (0.01)	-0.0044 (0.01)	0.0075 (0.02)	-0.0159 (0.04)	-0.0003 (0.01)	-0.0009 (0.009)	0.0036 (0.02)	-0.0250 (0.03)	
VC	0.0006 (0.07)	0.0006 (0.06)	-0.0005 (0.43)	-0.4603 (0.43)					
Female × VC	0.1110 (0.08)	0.1108 (0.13)	0.0668 (0.78)	0.9983 (0.78)					
Bank loans					-0.0025 (0.01)	0.0048 (0.01)	0.0473 (0.04)	0.0867 (0.05)	
Female × Bank loans					0.0152 (0.01)	-0.0022 (0.01)	-0.0277 (0.04)	-0.0484 (0.05)	
Startup & Founders characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Sector FE × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
R ²	0.473	0.561	0.085	0.149	0.381	0.464	0.083	0.135	
N	16,496	16,496	9,270	3,550	26,034	26,034	13,675	7,165	
Mean dep. var.	0.5994	0.2059	0.5141	0.4992	0.6436	0.5587	0.5566	0.5015	

Source: SINE survey, firm registry, and tax files. Sample: New firms founded in 2002, 2006, 2010, and 2014. This table uses OLS to analyze the performance of female-founded versus male-founded startups in male-dominated sectors (panel A) and female-dominated sectors (panel B). The dependent variables are the likelihood of surviving after 3 years (columns 1 and 5), after 5 years (columns 2 and 6), the sales growth between year 0 and year 3 (columns 3 and 7), and between year 0 and year 5 (columns 4 and 8). The main independent variables are the founder's gender, Female, and its interaction with VC financing (columns 1 and 5) and bank financing (columns 3 and 7), respectively. All models include the baseline founders and startups characteristics of Table 3, county, four-digit French SIC sector × cohort-year fixed effects. Clustered standard errors at the sector level are reported in parentheses. *p < .1, **p < .05, ***p < .01.

(b) Panel B: Female-dominated sectors

Table 10: Outcome test of equity-backed and bank-financed startups

8.11 Table 11: Gender gap and banks' supply-side effects

How to read it. Table 11 uses the PME survey to test whether female entrepreneurs experience different bank-loan application outcomes, guarantee requirements, or rejection reasons. **Economic**

message: The table supports the claim that banks do not treat female entrepreneurs differently in the same way equity investors do. Female founders are not significantly more likely to face guarantees or different rejection reasons. This helps explain why debt can partially substitute for equity.

Table 11
Gender gap and banks' supply-side effects

Panel A: Substitution effects between equity and debt

	Use of Financing in 2007 & 2010			Application to Financing in 2007 & 2010		
	VC (1)	Bank loans (2)	VC (3)	VC Application (4)	Bank Application (5)	VC Application (6)
<i>Female-founded</i>	-0.0108** (0.005)	-0.0212 (0.029)	0.0008 (0.004)	-0.0181** (0.007)	-0.0481 (0.032)	-0.0067 (0.008)
<i>Bank loans</i>			0.0377** (0.015)			0.0386** (0.015)
<i>Female × Bank loans</i>			-0.0407** (0.017)			-0.0399** (0.019)
Sector × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.283	0.159	0.294	0.267	0.167	0.275
Observations	743	743	743	743	743	743
Mean dep. var.	0.0188	0.3001	0.0188	0.0269	0.3351	0.0269

Panel B: Bank guarantees

	Any guarantees (1)	Personal guarantees (2)	Family (3)	Collective guarantee system (4)	Public (5)	Other (6)
<i>Female-founded</i>	-0.1349 (0.111)	-0.1285 (0.096)	0.0269 (0.040)	-0.0378 (0.045)	0.0048 (0.059)	-0.0208 (0.038)
Sector × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.206	0.244	0.122	0.207	0.282	0.278
Observations	213	230	230	230	230	230
Mean dep. var.	0.6526	0.4391	0.0130	0.0696	0.2043	0.0522

Panel C: Reasons for being rejected

	Poor rating (1)	Not enough equity (2)	Too much debt (3)	Insufficient collateral (4)	Too risky (5)	No reason given (6)
<i>Female-founded</i>	-0.0197 (0.064)	-0.0395 (0.070)	-0.0294 (0.029)	0.0341 (0.051)	-0.0170 (0.045)	0.0215 (0.039)
Sector × Cohort-year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.247	0.204	0.159	0.209	0.199	0.261
Observations	230	230	230	230	230	230
Mean dep. var.	0.1120	0.1317	0.0204	0.1087	0.0836	0.0565

Table 11: Gender gap and banks' supply-side effects

9 Detailed interpretation of the mechanisms

9.1 Why male-dominated sectors matter

Male-dominated sectors are central to the paper because they create a setting in which gender is likely to be salient. If an investor has a mental prototype of a successful founder in a sector, and that prototype is male, a female founder may be perceived as less representative even when her observable characteristics are strong. This is different from saying that all investors dislike women or that all women have weaker startups. The paper’s evidence points to a more contextual mechanism: the same female founder profile is evaluated differently depending on the gender type of the sector.

9.2 Why bank loans are a useful comparison group

The comparison between equity financing and bank debt is useful because these capital providers use different evaluation technologies. Banks focus on repayment capacity, creditworthiness, collateral, and tangible assets. Equity investors care more about exceptional upside and often rely on soft signals about the founder. The paper shows that women do not face the same negative gap in bank debt as they do in equity. This makes a pure demand-side story less plausible.

9.3 Why outcome tests matter

Outcome tests ask whether funded women perform better or worse after receiving funding. If women have to be better than men to obtain VC, then conditional on receiving VC, women should outperform. The paper finds precisely this in male-dominated sectors. The result is not perfect causal proof because VC selection is not randomized, but it is an important consistency check: a higher bar for women should generate exactly this performance pattern.

10 Potential limitations and how the paper addresses them

10.1 Demand for VC is hard to observe

The main limitation is that researchers do not observe every entrepreneur’s actual VC application decision in the SINE data. The paper addresses this by controlling for growth intentions, incorporation, innovation, serial entrepreneurship, startup capital, and motivations, and by restricting samples to founders who look likely to demand equity.

10.2 Unobserved quality remains possible

Even with rich controls, unobserved project quality could remain. The paper addresses this concern through Oster bounds, sector-gender heterogeneity, comparison with bank debt, and outcome tests. These do not fully eliminate the concern, but they make a simple omitted-quality explanation less convincing.

10.3 External validity

The paper studies France, where VC markets, labor markets, and public funding programs differ from those in the United States. However, the mechanisms are relevant more broadly because the paper’s main contribution is not a France-specific financing fact, but a general test of context-dependent stereotypes using unusually rich population-level data.

11 Conclusion and takeaway

Hebert (2025) shows that the gender gap in entrepreneurial equity financing is not only a pipeline problem or a founder-characteristics problem. Female entrepreneurs are less likely to use external equity even after controlling for rich founder and startup information, and the gap is concentrated in male-dominated and innovative sectors. Women with high-growth profiles appear to substitute bank debt for missing equity, but this is an incomplete substitute because debt is less suitable for risky, intangible, high-upside projects. The strongest evidence for stereotypes comes from the interaction of three results: the gap is sector-contextual, observable quality signals do not close it, and VC-backed female founders outperform in male-dominated sectors. The paper therefore argues that context-dependent stereotypes among equity investors distort entrepreneurial financing and may reduce innovation and growth by underfunding high-potential female-led ventures.